Computational Methods for Single-Cell RNA-Seq Analysis of *Arabidopsis thaliana*

Chapter 2 - WT vs pifq: pseudobulk DE, GO enrichment, heatmaps and network sections

SingleCell Pipeline

2026-07-15

Purpose

This Chapter 2 draft follows the structure of `workflow/capitulo2_pseudobulk_de.R`. The WT/pifq adaptation changes only the dataset-specific pieces: output folder `resultados_wt`, input object `pbmc_harmony_annotated.rds`, direct annotation column `celltype`, and the `WT_vs_pifq` contrast.

No grouped or curated labels are used here. This keeps Chapter 2 aligned with the direct bibliography annotation from Chapter 1.

Configuration

Configuration mirrors the original Chapter 2 script, with the WT/pifq result directory and the same function sources.

R

```r
PIPELINE_DIR <- "/workspace/SingleCell/workflow"
DATA_DIR     <- "/workspace/."
base_dir     <- file.path(DATA_DIR, "resultados_wt")

source(file.path(PIPELINE_DIR, "load_libraries.R"))
source(file.path(PIPELINE_DIR, "custom_seurat.R"))
source(file.path(PIPELINE_DIR, "ScRNA_Analysis_Functions.R"))

list2env(create_pipeline_dirs(base_dir), envir = .GlobalEnv)  # creates output
folders and loads their paths as variables

set.seed(1807)
```

Load - Annotated Object From Part 1

The input is the Chapter 1 bibliography-annotated object. The active annotation column for Chapter 2 is `celltype`.

R
```r
pbmc_harmony <- readRDS(file.path(dir_objects, "pbmc_harmony_annotated.rds"))
```

Section 13 - Cell-Type Subsets

One Seurat subset is created per direct `celltype` label. Object names are sanitized only for list names and filenames.

R

```r
pseudobulk_annot_col <- "celltype"

cell_type_subsets <- create_cell_type_subsets(pbmc_harmony, annot_col =
pseudobulk_annot_col)



message("\n✓ SECTION 13 COMPLETE: Cell-type subsets created")
```

Section 14 - Pseudo-Replicate Assignment

The same pseudo-replicate assignment structure is retained. The QC check uses `Mesophyll` as an existing label in the WT/pifq dataset.

```r
pseudobulk_conditions <- c("WT", "pifq")
n_pseudoreps          <- 3

cell_type_subsets_replicates <- assign_pseudoreplicates_batch(
  cell_type_subsets,
  pseudobulk_conditions = pseudobulk_conditions,
  n_pseudoreps = n_pseudoreps
)


table(cell_type_subsets_replicates$Mesophyll$condition)
table(cell_type_subsets_replicates$Mesophyll$replicate)
table(cell_type_subsets_replicates$Mesophyll$orig.ident)



message("\n✓ SECTION 14 COMPLETE: Pseudo-replicates assigned")
```

Section 15 - Pseudobulk Tables and DESeq2

The contrast is `WT_vs_pifq`. Positive log2FC values correspond to higher expression in pifq relative to WT.

R

```
comparisons <- list(
  list(conds = c("WT", "pifq"), tag = "WT_vs_pifq")
)

cell_types_to_analyze <- NULL

output_dir <- dir_06

deseq2_results <- run_pseudobulk_deseq2_analysis(
  cell_type_subsets_replicates = cell_type_subsets_replicates,
  comparisons = comparisons,
  output_dir = output_dir,
  cell_types = cell_types_to_analyze,
  pseudobulk_dir = file.path(dir_objects, "pseudobulk_replicas")
)



message("\n✓ SECTION 15 COMPLETE: Pseudobulk aggregation and DESeq2 complete")
```

Section 16 - Volcano Plots

R

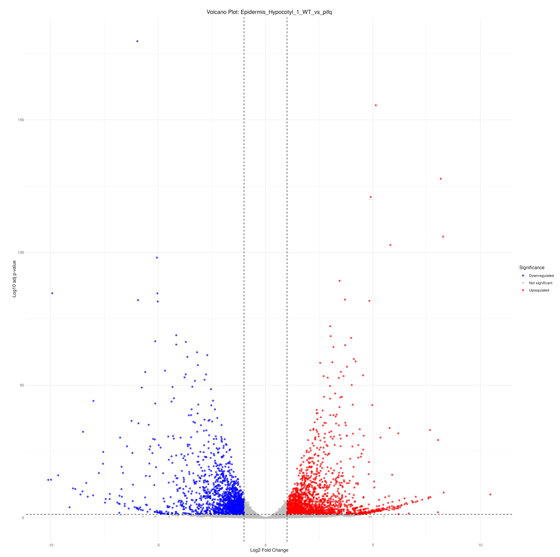
```
volcano_tag <- "WT_vs_pifq"
padj_cut    <- 0.05
lfc_cut     <- 1

render_volcano_plots(
  results_dir = file.path(dir_06, volcano_tag),
  output_dir  = file.path(dir_06, volcano_tag, "volcano"),
  pdf_name    = paste0("VolcanoPlots_", volcano_tag, ".pdf"),
  padj_cut    = padj_cut,
  lfc_cut     = lfc_cut
)



message("\n✓ SECTION 16 COMPLETE: Volcano plots saved")
```

Representative volcano output: Mesophyll.



Representative volcano output: Epidermis Hypocotyl.1.

Section 17 - Differential Gene Tables

R

```r
diff_prefix <- paste0("diff_table_", volcano_tag)

diff_tables <- build_differential_tables(
  results_dir = file.path(dir_06, volcano_tag),
  output_dir  = file.path(dir_06, volcano_tag),
  padj_cut    = padj_cut,
  lfc_cut     = lfc_cut,
  prefix      = diff_prefix
)



message("\n✓ SECTION 17 COMPLETE: Differential gene tables saved")
```

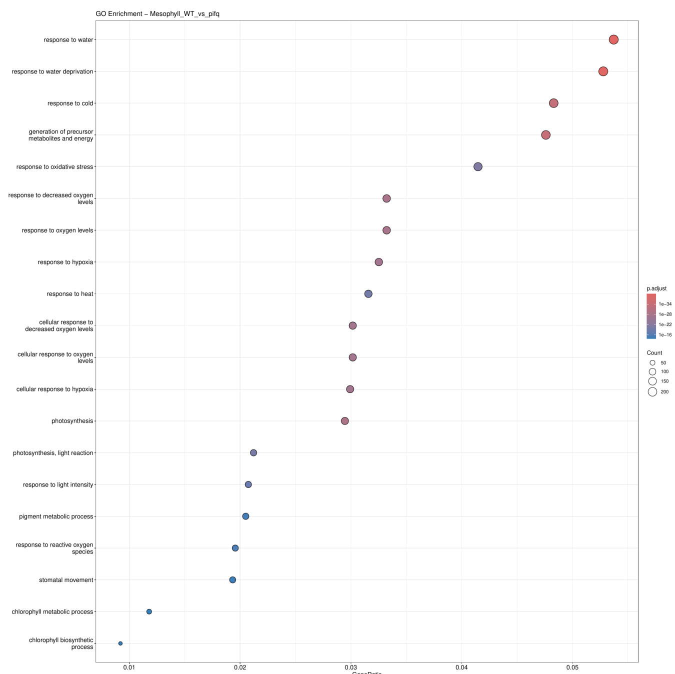
Section 18 - GO Enrichment

R

```
go_space    <- "BP"
go_orgdb    <- org.At.tair.db
go_keytype  <- "TAIR"
padj_cutoff <- 0.05

go_results <- run_go_enrichment_for_contrast(
  results_dir  = file.path(dir_06, volcano_tag),
  output_dir   = file.path(dir_07, volcano_tag),
  orgdb        = go_orgdb,
  keytype      = go_keytype,
  go_space     = go_space,
  padj_cutoff  = padj_cutoff,
  contrast_tag = volcano_tag
)


message("\n✓ SECTION 18 COMPLETE: GO enrichment complete")
```



Representative GO biological-process bubble plot for Mesophyll.

Section 19 - Log2FC Heatmap

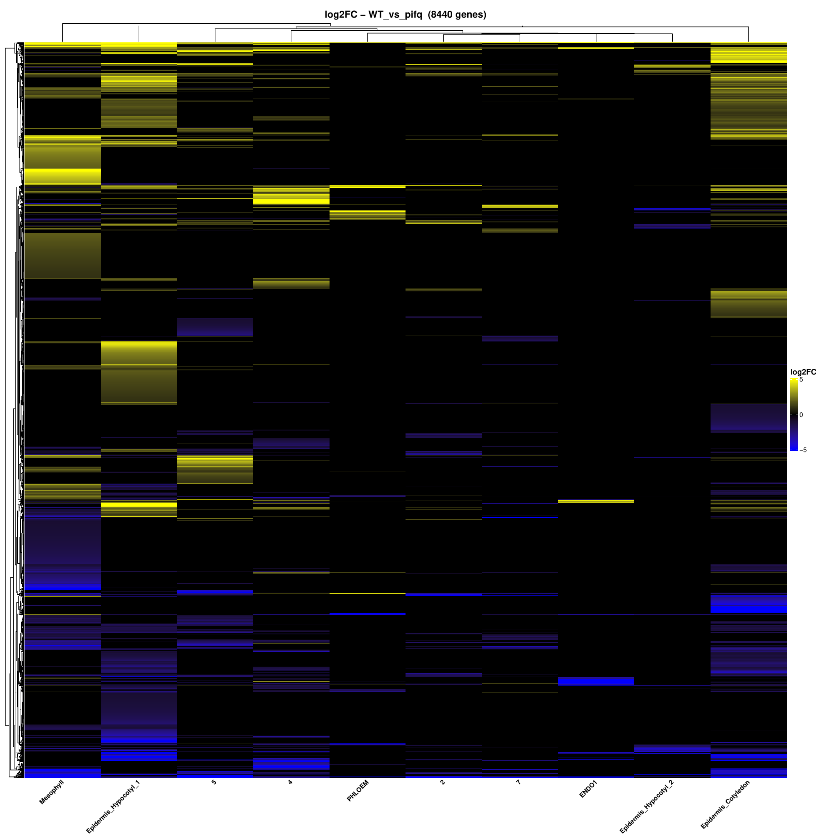
R

```r
heatmap_limits <- c(-5, 5)

build_logfc_heatmap(
  logfc_table  = diff_tables$logfc,
  contrast_tag = volcano_tag,
  output_dir   = file.path(dir_06, volcano_tag),
  limits       = heatmap_limits
)



message("\n✓ SECTION 19 COMPLETE: Log2FC heatmap saved")
```



log2FC heatmap produced by Section 19.

Section 20 - Coexpression Network

The hdWGCNA section is preserved from the original Chapter 2 structure. It uses the significant-gene table generated in Section 19.

R

```
wgcna_name  <- "unified"
n_metacells     <- 50
soft_power      <- NULL
min_module_size <- 20
deep_split      <- 2

run_unified_hdwgcna(
  seurat_obj      = pbmc_harmony,
  de_table_path   = file.path(dir_06, volcano_tag,
"tabla_log2FC_fc1_padj_005.tsv"),
  output_dir      = dir_08,
  annot_col       = pseudobulk_annot_col,
  sample_col      = "orig.ident",
  wgcna_name      = wgcna_name,
  n_metacells     = n_metacells,
  soft_power      = soft_power,
  min_module_size = min_module_size,
  deep_split      = deep_split
)

message("\n\u2713 SECTION 20 COMPLETE: hdWGCNA co-expression networks saved")
```

Section 21 - Network Export and Visualization

R

```
tom_threshold <- 0.2
n_hub_label   <- 5

plot_hdwgcna_network(
  hdwgcna_dir   = dir_08,
  output_dir    = dir_08,
  tom_threshold = tom_threshold,
  n_hub_label   = n_hub_label
)

message("\n✓ SECTION 21 COMPLETE: Network files and plots saved")
```

Section 21b - TF Coexpression Network

R

```r
tom_threshold_tf <- 0.2
n_hub_label_tf   <- 15

edges_all <- read.table(file.path(dir_08, "edges_unified.tsv"),
                        header=TRUE, sep="\t")
tfs       <- trimws(readLines(file.path(DATA_DIR, "SingleCell", "data",
"AtTFDB_loci.txt")))
de_mat    <- read.table(
  file.path(dir_06, volcano_tag, "tabla_log2FC_fc1_padj_005.tsv"),
  header=TRUE, sep="\t", row.names=1, check.names=FALSE)

edges_tf <- edges_all[edges_all$weight >= tom_threshold_tf, ]
net_tf   <- build_tf_network(edges_tf, tfs, de_mat)
plot_tf_de_network(net_tf,
                   output_dir    = dir_08,
                   layout        = "graphopt",
                   n_hub_label   = n_hub_label_tf,
                   contrast_tag  = volcano_tag,
                   output_pdf    = "network_tf_DE_direction.pdf",
                   output_width  = 22,
                   output_height = 22)

message("\n✓ SECTION 21b COMPLETE: TF network saved to
network_tf_DE_direction.pdf")
```